

CSCE 548 – Project 4 Deployment Document

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Overview

This project is an n-tier web application consisting of:

- Data Layer
 - Business Layer
 - Service Layer
 - Client Layer
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Prerequisites

To run this project, the following software is required:

- Python 3.x
 - pip (Python package manager)
 - Node.js
 - npm
 - Git
 - Web browser
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Step 1 – Download the Project

1. Go to the GitHub repository:
<https://github.com/anniekoshy1/CSCE-548>
2. Click “Code” → “Download ZIP”
3. Extract the ZIP file
4. Open the folder in VS Code or terminal

Step 2 – Install Backend Dependencies

In terminal:

```
pip install flask flask-cors
```

Step 3 – Initialize the Database

Run:

```
python init_db.py
```

This creates the SQLite database file (homework.db)

Step 4 – Start the Backend Service

Run:

```
python service.py
```

The API will run at:

```
http://127.0.0.1:5000
```

Step 5 – Start the Frontend

In a new terminal:

```
cd client  
npm install  
npm run dev
```

Open the app in browser:

<http://localhost:5173>

Step 6 – Verify the System Works

You should see the client interface with:

- Users
- Courses
- Assignments

Test the following:

- Create a user
- View all users
- View a single user
- Update a user

If data appears and updates correctly, the system is working.

System Testing

The following operations were tested:

- Insert (Create user)
- Get all users
- Get single user
- Update user

Screenshots of these tests are included in the document.

Conclusion

The application successfully demonstrates full functionality across all layers, including data persistence and user interaction through the client interface.